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AI CAREERS FOR WOMEN

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Artificial Intelligence in the Contemporary World

Student Feedback & Teaching Improvement Analyzer

Project Domain

Educational Analytics & NLP

[Excel / Power BI with Copilot / AI Chatbot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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1. Introduction

Student feedback is a critical component of quality assurance in higher education. Educational institutions collect feedback every semester to evaluate teaching effectiveness, course structure, and learning experience. However, most institutions store feedback data in spreadsheets without systematic analysis.

This project proposes a Student Feedback & Teaching Improvement Analyzer that uses Excel datasets, analytical charts, dashboards, and AI-based insights to convert feedback data into actionable recommendations for improving teaching quality.

2. Problem Statement

Educational institutions collect large volumes of student feedback data through surveys, forms, and evaluations. However, this data is often unstructured, inconsistent, and difficult to analyze effectively. As a result, valuable insights into teaching quality, student satisfaction, and areas for improvement remain underutilized.

With the rise of Artificial Intelligence and Generative AI tools such as Microsoft Copilot and AI Chatbots, there is an opportunity to transform raw feedback into actionable insights. Yet, most educators and administrators lack the skills to clean, analyze, and visualize this data or to leverage AI-driven prompt engineering for meaningful interpretation.

This project addresses these challenges by applying data cleaning, analysis, visualization, and AI integration techniques to student feedback datasets. It will demonstrate how structured workflows combined with AI assistance can help institutions improve teaching quality, enhance student learning experiences, and support evidence-based decision-making.

Students should include in their write-up:

- Background of student feedback systems and challenges in analyzing them
- Why improving teaching quality through feedback analysis is significant
- The specific inefficiency (manual, time-consuming analysis) the project aims to address
- Who benefits (students, faculty, administrators) and the value delivered (better teaching outcomes, improved satisfaction)

3. Objectives

- Collect and explore student feedback datasets to understand structure, fields, and quality issues.
- Clean and preprocess the data (remove duplicates, handle missing values, standardize formats).
- Apply domain-relevant analysis using Excel formulas, pivot tables, or Power BI measures.
- Build interactive dashboards or structured reports that highlight teaching strengths and areas needing improvement.
- Integrate AI tools (Microsoft Copilot / AI Chatbot) to generate summaries, recommendations, and automated insights.
- Document the complete process, upload files to GitHub, and record a demonstration video.
- (Project-specific objective) Create a **prompt library** for AI-based feedback summarization and teaching improvement suggestions.

4. Methodology / Implementation Plan

#	Phase	Description
1	**Data Collection & Exploration**	Obtain student feedback datasets (CSV/Excel) from institutional sources or public repositories. Understand field names, data types, and row count.
2	**Data Cleaning & Preprocessing**	Remove duplicates, handle missing values, fix incorrect data types, and standardize formats using Excel/Power Query. For AI prompt projects, design structured prompts for feedback analysis.
3	**Analysis & AI Integration**	Apply pivot tables, Excel formulas (SUMIF, COUNTIF, VLOOKUP), or Power BI DAX measures to derive insights. Use Copilot/Chatbot to generate narratives and recommendations.

4	**Visualization / Output**	Build dashboards with charts, KPI cards, and slicers in Power BI or Excel. Document AI prompts and outputs.
5	**Documentation & Submission**	Write the project report, upload files to GitHub, and record a video demonstration. Submit GitHub link, video link, and dashboard link.

5. Resources and Timeline

A. Hardware & Software Requirements

- Laptop/PC (Intel Core i3 or higher, 8 GB RAM recommended)
- Windows 10/11 or macOS
- Microsoft Excel (2019 or later)
- Power BI Desktop (latest version)
- Microsoft Copilot (via Microsoft 365 subscription)
- AI Chatbot (ChatGPT / Gemini / Copilot Web)
- Internet connection & browser (Edge/Chrome)

B. Dataset

- Student feedback dataset (institutional or simulated)
- Source link if publicly available (e.g., Kaggle, UCI Repository)
- For AI prompt projects: description of use case and reference content

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*

3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

6. Expected Outcomes / Conclusion

- A clean, validated student feedback dataset with documented preprocessing steps.
- An interactive dashboard/report highlighting teaching strengths and improvement areas.
- Demonstrated use of AI tools (Copilot/Chatbot) for generating insights and recommendations.
- A GitHub repository with project files, report, and video demonstration.
- Project-specific outcome: A **prompt library** for AI-driven teaching improvement analysis.

7. References

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026.
- LinkedIn Learning. AI and Data Analytics Courses. 2026.
- Microsoft Corporation. Excel Documentation.
- Microsoft Corporation. Power BI Documentation.
- Microsoft Corporation. Microsoft Copilot.
- Kaggle Inc. Open Datasets.
- [Add project-specific references: research papers, GitHub links, YouTube tutorials, blogs, etc.]